
1.1 Basics of network

→ Computer Networking is the practice of connecting computers togetherto enable communication and data exchange between them. In general, Computer Network is a collection of two or more computers. Ithelps users to communicate more easily.

→ A computer network is a group of two or more interconnected computer systems. You can establish a network connection using either cable or wireless media.

→ Every network involves hardware and software that connects computers and tools.

→ The working of Computer Networks can be simply defined as rules or protocols which help in sending and receiving data via the links which allow Computer networks to communicate.

→ Each device has an IP Address, that helps in identifying a device.

- ## Basic Terminologies of Computer Networks

- **Network:** A network is a collection of computers and devices thatare connected together to enable communication and data exchange.
- **Nodes:** Nodes are devices that are connected to a network. These can include computers, Servers, Printers, Routers, Switches, and other devices.

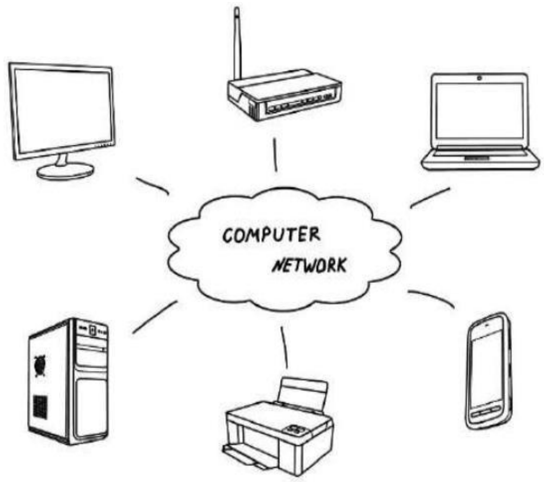
- **Protocol:** A protocol is a set of rules and standards that govern how data is transmitted over a network. Examples of protocols include TCP/IP, HTTP, and FTP.

- **Topology:** Network topology refers to the physical and logical arrangement of nodes on a network. The common network topologies include bus, star, ring, mesh, and tree.

- **IP Address**: An IP address is a unique numerical identifier that is assigned to every device on a network. IP addresses are used to identify devices and enable communication between them.

- **DNS:** The Domain Name System (DNS) is a protocol that is used to translate human-readable domain names (such as www.google.com) into IP addresses that computers can understand.

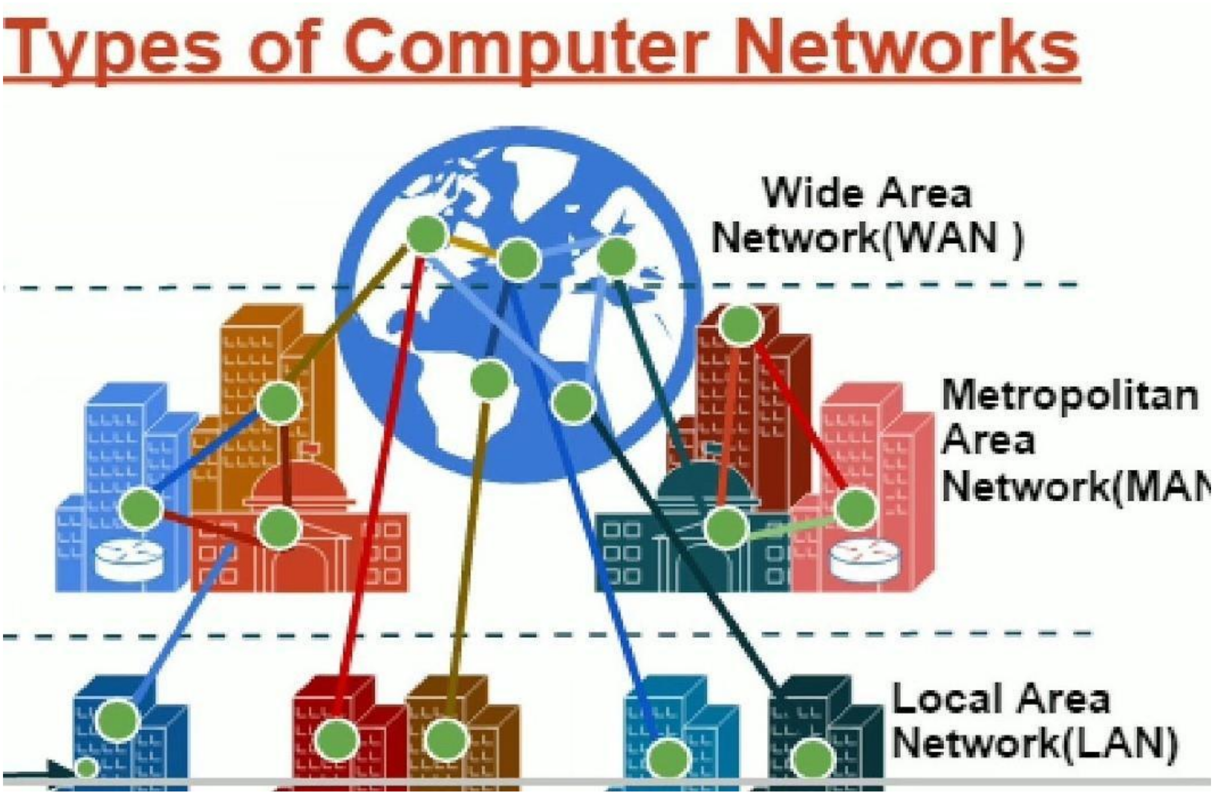
- **Firewall:** A firewall is a security device that is used to monitor and control incoming and outgoing network traffic. Firewalls are used to protect networks from unauthorized access and other security threats.



→ A computer network is a system in which multiple computersare connected to each other to share information and resources.

→ The physical connection between networked computing devices is established using either cable media or wirelessmedia.

→The best-known computer network is the internet.



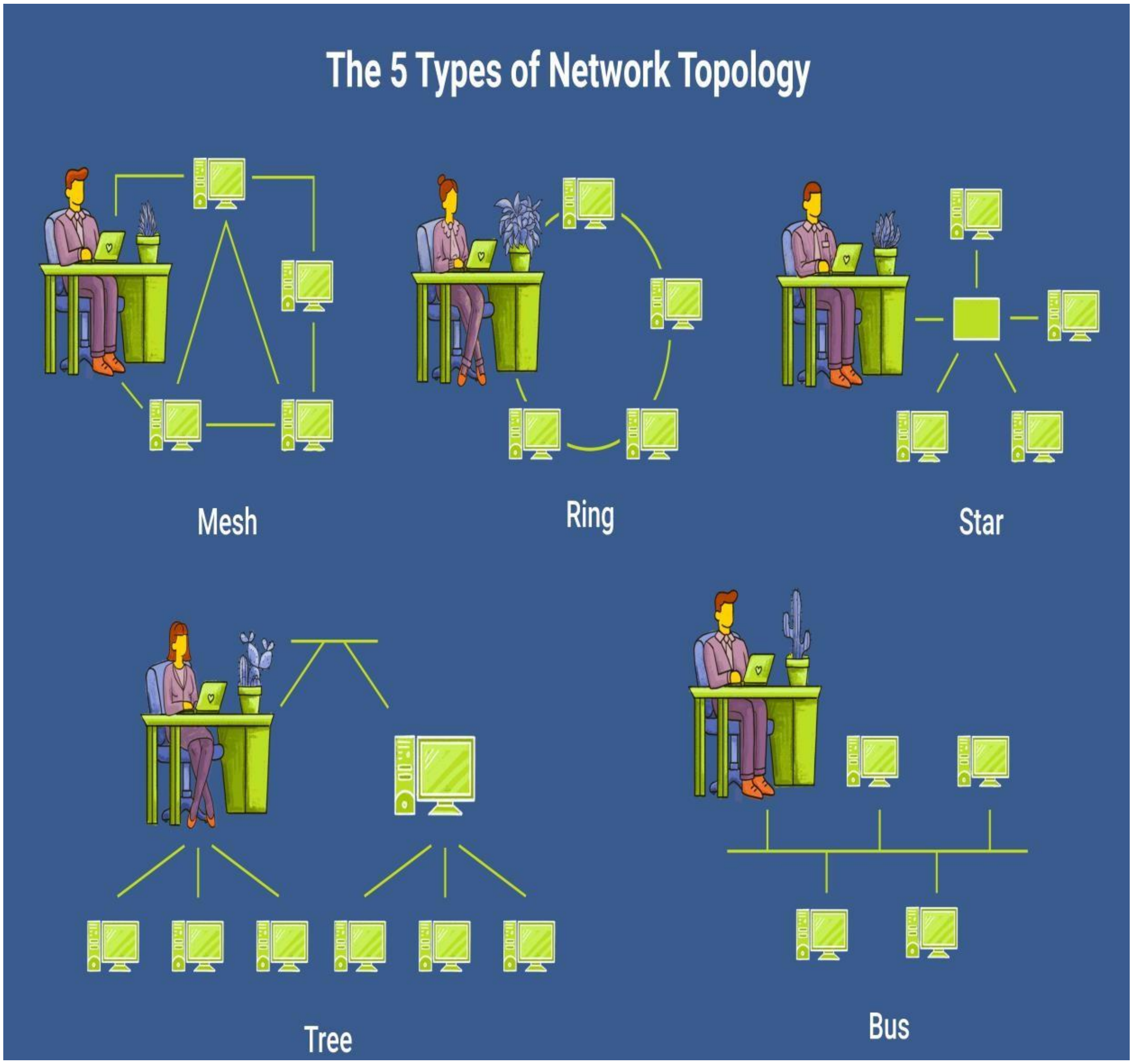
1.1.2 Different topologies (Bus, ring, star, mesh, tree)

→ Topology defines the structure of the network of how all the componentsare interconnected to each other.

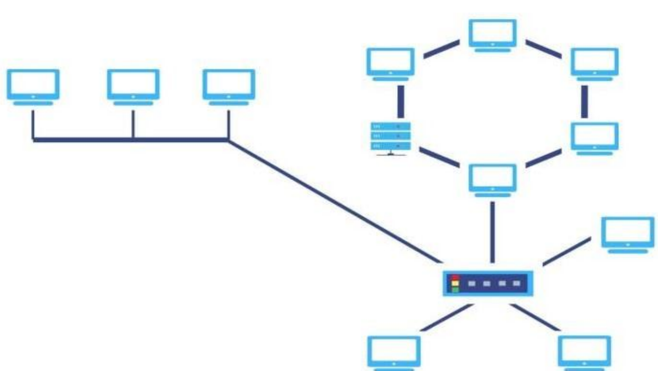
→ There are two types of topology: physical and logical topology.

→ Physical topology is the geometric representation of all the nodes in anetwork.

→ There are six types of network topology which are Bus Topology, RingTopology, Tree Topology, Star Topology, Mesh Topology, and Hybrid Topology.



6) Hybrid Topology

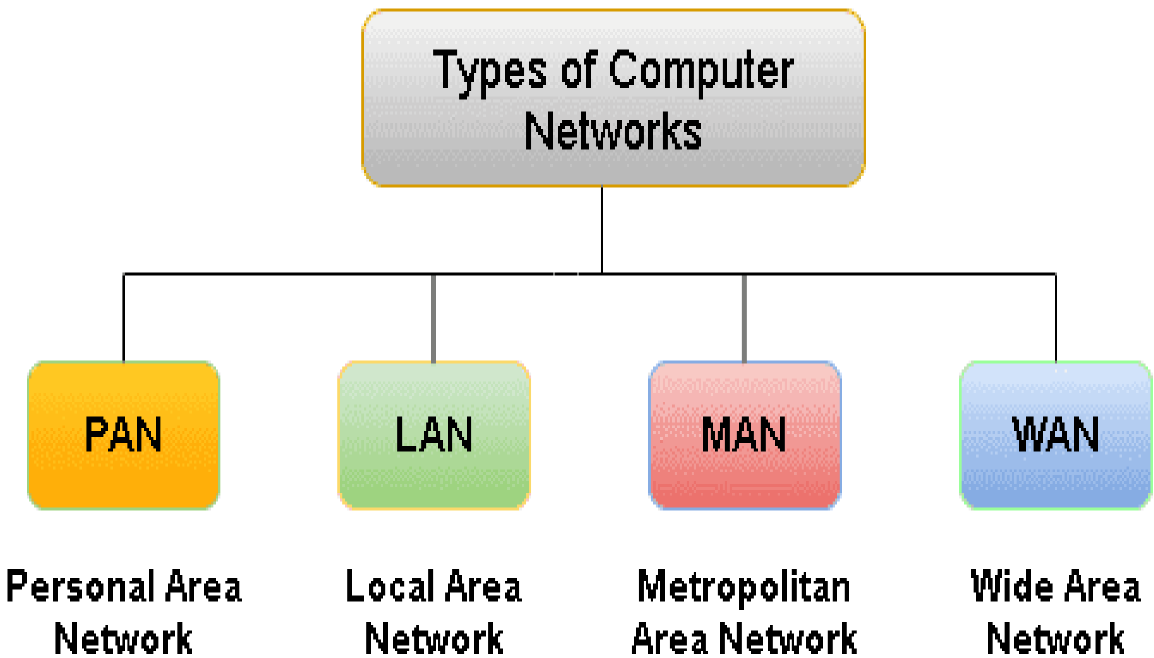


❖ Types of computer network?

→A computer network is a system in which multiple computers are connected to each other to share information and resources.

→The physical connection between networked computing devices is established using either cable media or wireless media.

→**According to the communication requirements, multiple types of network connections are available. The most basic type of network classification depends on the network's geographical coverage.**



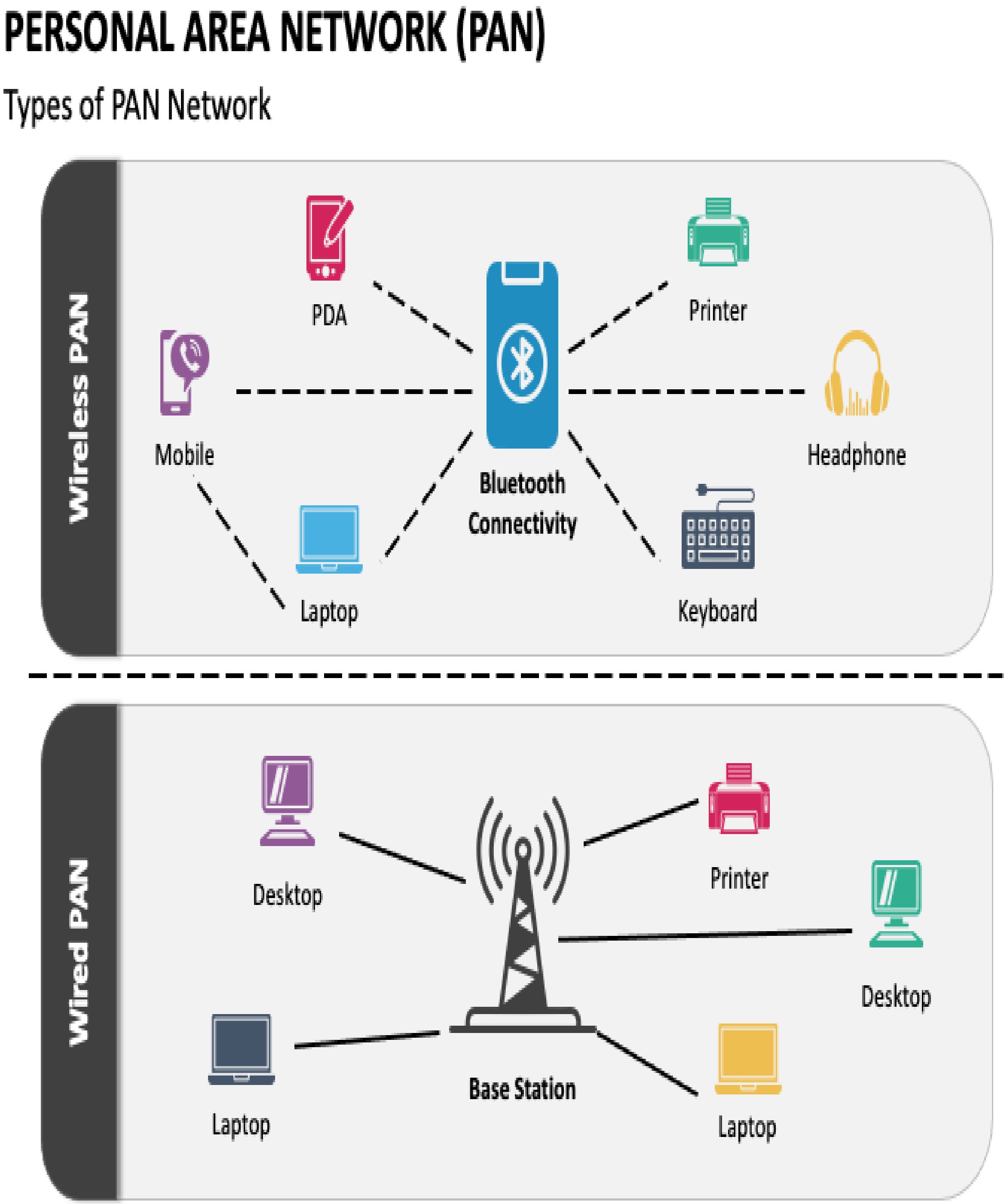
1) PAN(Personal area network)

→ Personal Area Network is a network arranged within an individual person.

→ Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.

→ Personal Area Network **covers an area of 30 feet**.

→ Personal computer devices that are used to develop the personal area network are the **laptop, mobile phones, media player and play stations.**



2) LAN(Local area network)

→it is privately-owned networks within a single building or campus of up to a few kilomiters in size.

→They are widely used to connect personal computers and workstation in company offices and factories to share resources (e.g., printers) and exchange information.

→In lan, all the machines are connected to a single cable.

→**The internet speed of LAN is very high, i.e., 1000 Mbps.**

→They transfer data at high speeds. The high transmission rate is possible in LAN because of the short distance between various computer networks.

→They exist in a limited geographical area.

→**Coverage range Between 100 to 1000 meters.**

Advantages

→ Lan transfers data at high speed.

→ Lan technology is generally less expensive.

3) MAN(Metropolitan area network)

→ MAN is a larger version of LAN which covers an area that is larger than the covered by LAN but smaller than the area covered by WAN.

→ MAN connects two or more LANs.

→ **Coverage range between 10 to 100 kilometers.**

→**The speed of MAN is moderate, i.e., 44-155 Mbps.**

→ A metropolitan area network or MAN covers a city. The best-known example of a MAN is the cable television network available in many cities.

→ There are various examples of a MAN network that we can observe daily. Some of those are:

→ Cable TV network
→ Telephone networks

→Advantages of MAN:-

→ It can send data in both directions at the same time.

→ Metropolitan Area Network allows people to connect LANs.

→ Offers greater security than a WAN.

→Disadvantage of MAN:-

→ The data rate is slow in a Metropolitan Area Network compared to LAN.
→ Compared to LAN, more cable is required to set up a Metropolitan Area Network.
→ Because this network comprises (covered) multiple LANs, it is difficult to keep hackers out.

4) WAN(Wide Area network)

→ WAN spans a large geographical area, often a country or region.

→ WAN links different metropolitan's countries and national boundaries thereby enabling easy communication.

→ The speed of WAN is **relatively less** than MAN and LAN, i.e., **150 Mbps.**

→ It may be located entirely within a state or a country or it may be interconnected around the world.

→ The communication between different users of WAN is established using leased telephone lines or satellite links and similar channels.

→ WAN or Wide Area Network is a computer network that extends over a large geographical area, although it might be confined within the bounds of a state or country.

→ WAN has a **range of above 50 km**. A WAN could be a connection of LAN connecting to other LANs via telephone lines and radio waves and may be limited to an enterprise …

2) Explain Topology in brief?

→ Topology defines the structure of the network of how all the components are interconnected to each other.

→ There are two types of topology: physical and logical topology.

→ Physical topology is the geometric representation of all the nodes in a network.

→ There are six types of network topology which are Bus Topology, Ring Topology, Tree Topology, Star Topology, Mesh Topology, and Hybrid Topology.

1) Bus Topology

→ Bus topology is a type of network topology in which all the devices are connected to a single cable which is called the backbone of the network.

→ Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.

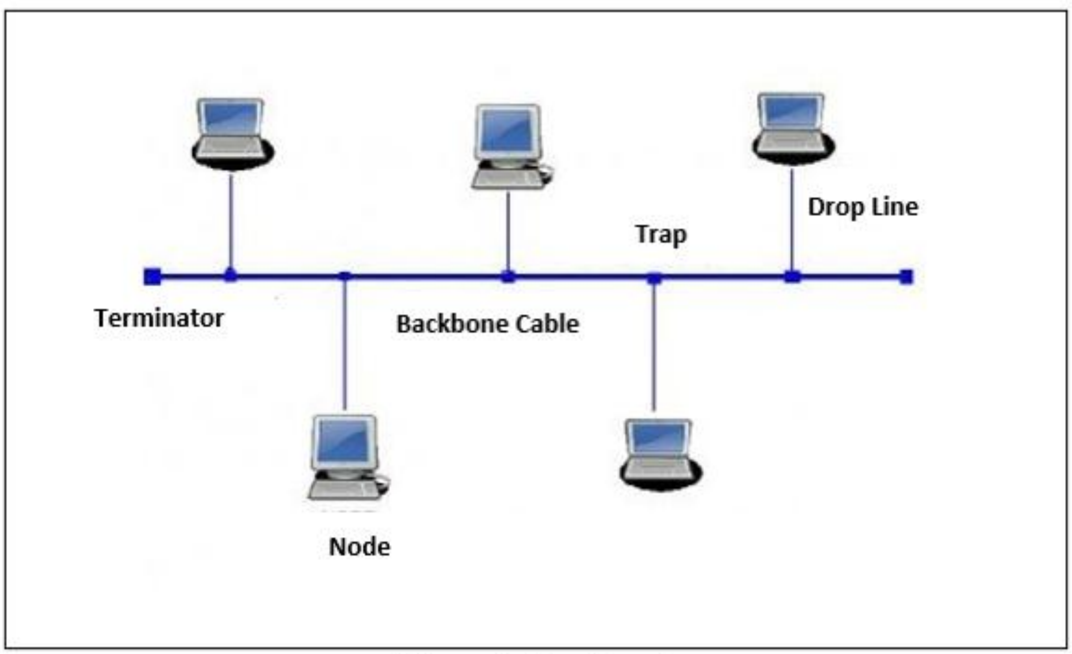


Figure 1: Bus Topology

→ It consists of a terminator at each end of the cable.

→ The network cable is responsible for the communication between the devices and when the data reaches the end of the cable it is removed by the terminator from the data line.

→ It is the easiest network topology when the devices are to be connected linearly.

→ The configuration of a bus topology is quite simpler as compared to other topologies.

→ The backbone cable is considered as a **"single lane"** through which the message is broadcast to all the stations.

→ **In a Bus topology, the failure of the network cable will cause the whole network to fail.**

→ In a Bus topology the data is transmitted slower.

- Ex:- The bus topology can be used for the further incorporate I/O devices like printers, scanners or others in a network at home/workplace.

⇒ **<u>Advantages :-</u>**

→ It is cost effective (cheaper).

→ Cable required is least compared to other network topology.

→ Used in small networks.

→ It is easy to understand.

→ Easy to expand joining two cables together.

⇒ **<u>Disadvantages:-</u>**

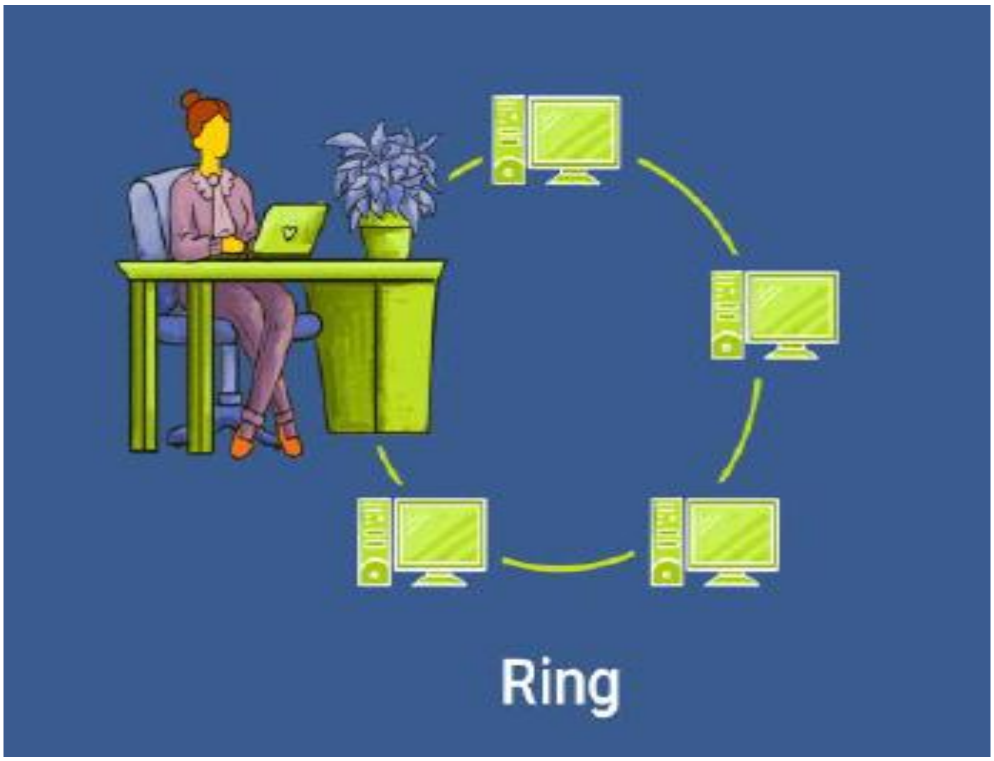
→ Cable fail then the whole network fails.

→ If network traffic is heavy or nodes are more the performance of the network decreases.

→ Cable has a limited length.

2) Ring Topology

→ It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first.



→ Exactly two neighbors for each device.

→ In this data flows in one direction.

→ Each node strongly connected to each other.

→ A number of repeaters are used and the transmission is unidirectional.

⇒ Advantages:

→ Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.

→ Cheap to install and expand.

⇒ **Disadvantages:**

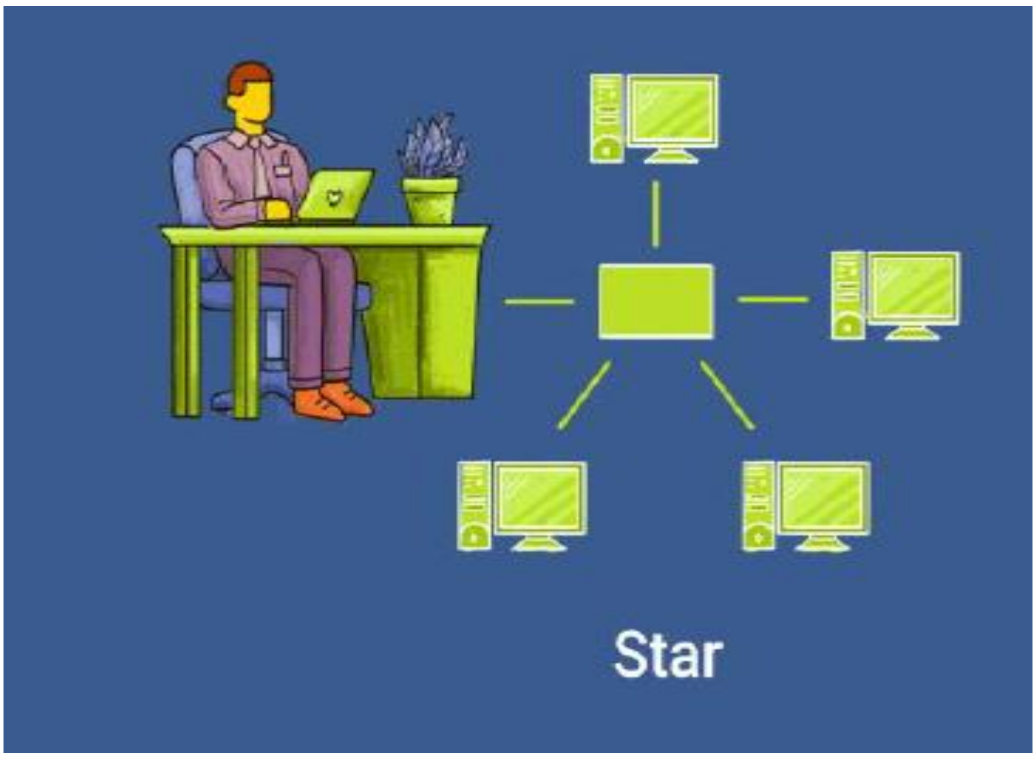
→ Troubleshooting is difficult in a ring topology.

→ Adding or deleting the computers disturbs the network activity.

→ Failure of one computer disturbs the whole network.

3) Star Topology

→ In this type of topology, all the computers are connected to a single hub through a cable.



→ This hub is the central node and all others nodes are connected to the central node.

→ Every node has its own dedicated connection to the hub.

→ Acts as a repeater for data flow.

→ Can be used with twisted pair, Optical Fiber or coaxial cable.

⇒Advantages:

→Fast performance with few nodes and low network traffic.

→Hub can be upgraded easily.

→Easy to troubleshoot.

→Easy to set up and modify.

→Only that node is affected which has failed rest of the nodes can work smoothly.

⇒Disadvantages:

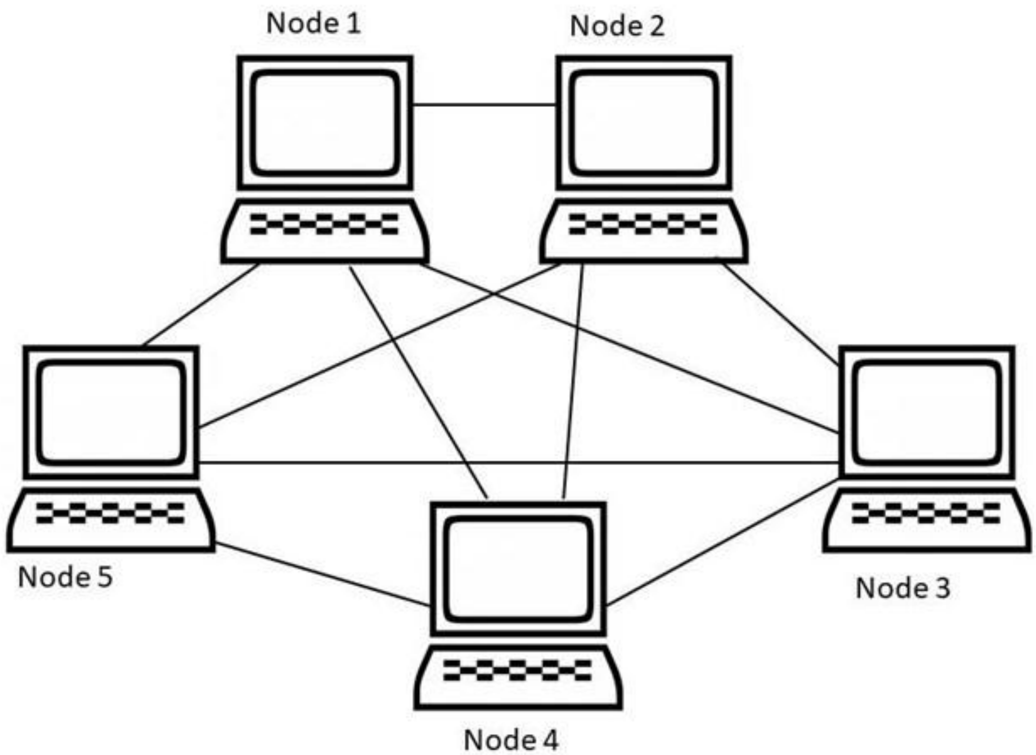
→Cost of installation is high.

→Expensive to use.

→If the hub is affected then the whole network is stopped because all the nodes depend on the hub.

4) Mesh Topology

→ It is a point to point connection to other nodes or devices.
→ Traffic is carried only between two devices or nodes to which it is connected.



→ The mesh topology has a unique network design in which each computer on the network connects to every other. It is develops a P2P (point-to-point) connection between all the devices of the network.

→ It offers a high level of redundancy, so even if one network cable fails, still data has an alternative path to reach its destination.

❖ Advantages

- The network can be expanded without disrupting current users.
- Each connection can carry its own data load.
- Need extra capable compare with other LAN topologies.

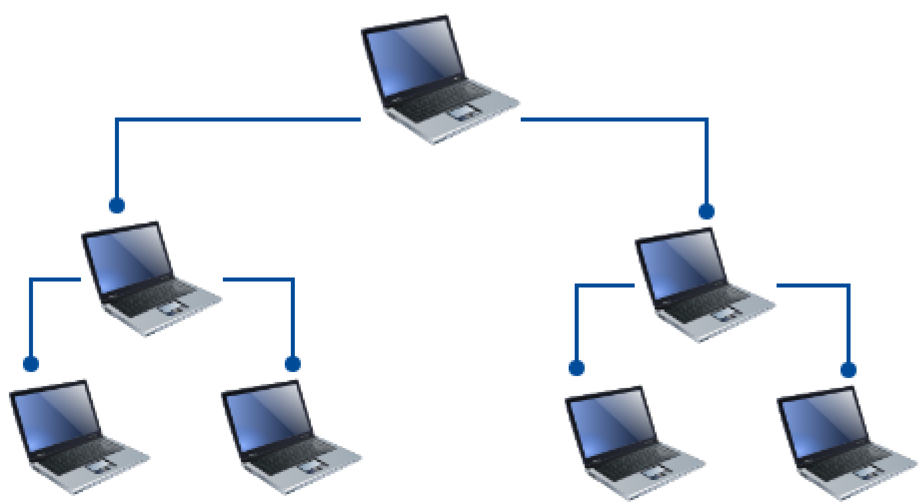
- No traffic problem as nodes has dedicated links.
- It has multiple links, so if any single route is blocked, then other routes should be used for data communication.
- Every system has its privacy and security.

Disadvantages

- Installation is complex because every node is connected to every node.
- It is expensive due to the use of more cables.
- Complicated implementation.

5) Tree topology

→ Tree topology is a very common network which is similar to a bus and star topology.



Tree Topology

→ It has a root node and all other nodes are connected to it forming a hierarchy.

→ It is also called hierarchical topology.

→ It should at least have three levels to the hierarchy.

→ Used in wide area network.

Advantages

- Failure of one node never affects the rest of the network.
- Detection of error is an easy process
- It is easy to manage and maintain

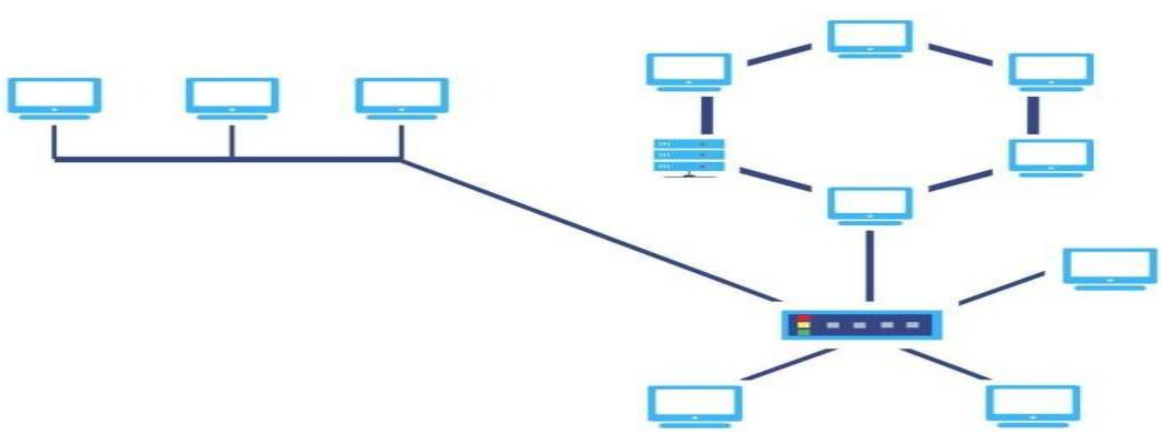
Disadvantage

→ Costly.
→ It is heavily cabled topology
→ If more nodes are added, then its maintenance is difficult
→ If the hub or concentrator fails, attached nodes are also disabled.
→ Central hub fails the network fails.

6) Hybrid topology

→ A network structure whose design contains more than one topology is said to be hybrid topology.

.



→ For example, if in an office in one department ring topology is used and in another star, topology is used, connecting these topologies will result in hybrid topology (ring topology and star topology).

Advantages

→ We can choose the topology based on the requirement.

→ It is scalable so you can increase your network size.

Disadvantages

→ Fault detection is difficult.

→ Installation is difficult.

→ Complex in design.

→ It is one of the costliest processes.

❖ What is internet?

→ The term internet has been derived from two terms, **interconnection** and **network**. A network is simply a group of computers that are connected together for sharing information, resources & communication. A Network several such networks has been joined together across the world to form internet.

→ Thus, Internet is a network of networks.

❖ What is an Intranet?

→ A part of the network, but controlled and used by a private organization.

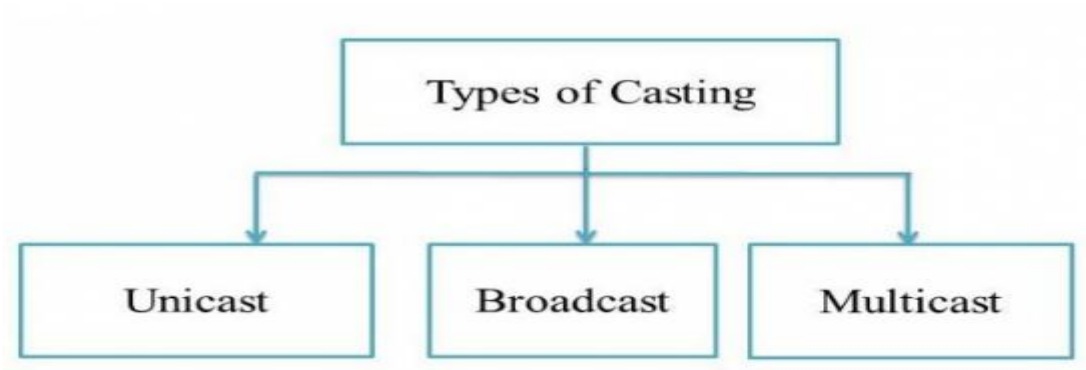
→ Intranet has restrictions and can support only fewer users.

→ Hence, only limited data can be shared over it.

Internet	intranet
The Internet is a global network that connects millions of devices and computers worldwide.	An intranet is a private network that connects devices and computers within an organization.
On the internet, there are multiple users.	On an intranet, there are limited users.
Internet is unsafe.	Intranet is safe.
On the internet, there is more number of visitors.	In the intranet, there is less number of visitors.
Internet is a public network.	Intranet is a private network.
Anyone can access the Internet.	In this, anyone can't access the Intranet.
The Internet provides unlimited information.	Intranet provides limited information.
Using Social media on your phone or researching resources via Google.	A company used to communicate internally with its employees and share information

❖ Explain unicast multicast and broadcast?

→ **Casting** in computer networks means transmitting data (stream of packets) over a network. Following are the different types of casting used in networking



❖ Unicast

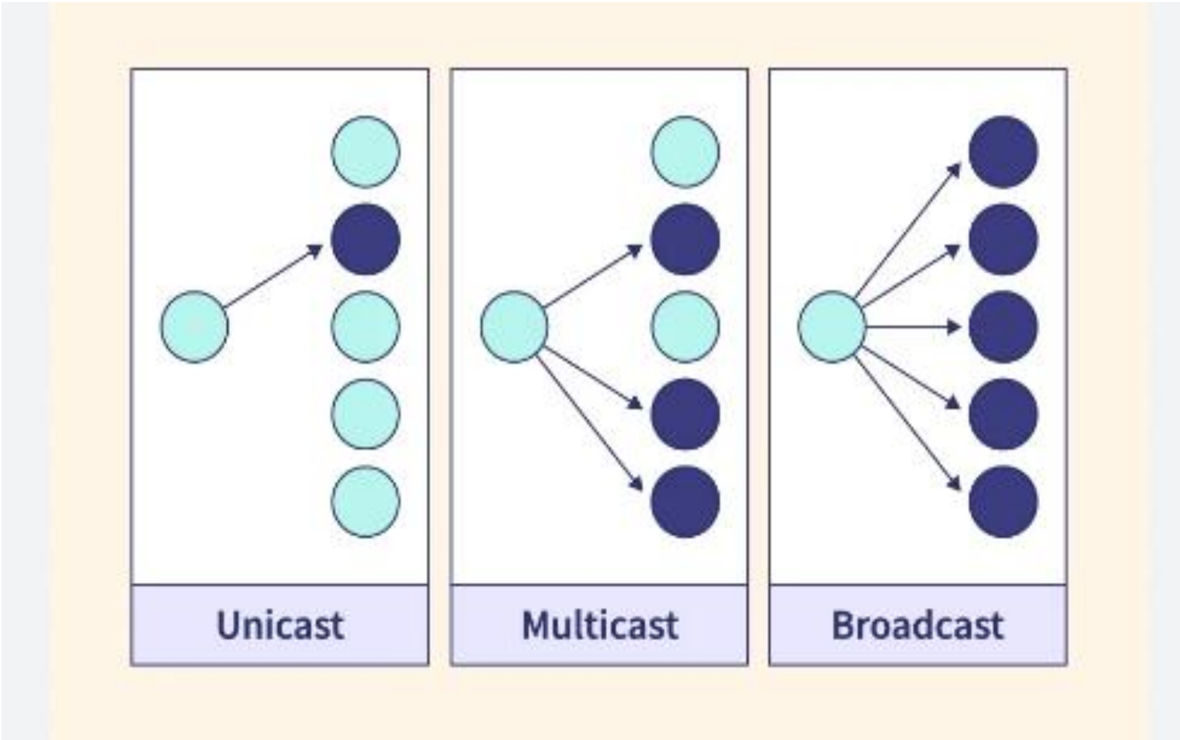
→ Unicast information transfer is when a single sender transmits data to a single recipient. It is a one-to-one type of transmission.

❖ Multicast

→ Multiple senders and recipients participate in the process of data transfer in Multicasting. It allows servers to direct a single copy of data streams to the hosts that requested them.

❖ Broadcast

→ Broadcast data transfer occurs when one sender transmits data to multiple recipients at any given time.



Unicast	Multicast	broadcast
There is only one receiver and one sender.	There are multiple receivers and multiple senders.	There are multiple receivers and one sender.
It is a one-to-one type of data transfer.	It is a many-to-many type of data transfer.	It is a one-to-many type of data transfer.
It is very helpful when a single sender transmits data to a single recipient.	These are helpful in the stock exchange, multimedia delivery, etc.	Broadcasting is mainly helpful for audio and video distribution by television networks.
Generates the least amount of network traffic	Generates moderate network traffic	Generates the most amount of network traffic
More secure because data is sent to a specific recipient	Moderately secure because data is sent to a specific group of devices	Less secure because data is sent to all devices in the network